VietProfs: A Public Directory of the Vietnamese Academic Diaspora

Purpose, Eligibility, and Design, with a Technical Report on Keeping It Accurate

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August 2026

Abstract

The Vietnamese academic diaspora is large and growing, yet no public, searchable record of it exists. Individual scholars are discoverable one at a time on their own university pages, but there is no shared directory that lets a student, journalist, organizer, or fellow academic find Vietnamese and Vietnamese-diaspora faculty across institutions, countries, and fields. We built VietProfs to fill this gap: a public directory intended to stay accurate as a living reference rather than a one-time listing. Each entry pairs a portrait and Vietnamese-diacritic name with verified appointment, education, and honors information, browsable on a dynamic web interface with easy search, filtering, and sorting capabilities. The directory is designed to be a living document that reflects the current state of the Vietnamese academic diaspora.

This paper, consisting of two parts, focuses on both the directory itself and the engineering challenges (and solutions) of creating and maintaining the data and the public interface. Part I describes VietProfs itself: its purpose and intended uses, exactly who is eligible to appear and why, how its 17 fields of study relate to standard U.S. federal classifications used by the NSF and NIH, what information is shown about each person and the design reasoning behind it, and what the current roster reveals. Part II is an experience report on the harder problem that keeping the directory accurate exposed, using VietProfs as the deployed instance. Its workflow combines bounded search tasks, agent-assisted candidate generation and revalidation, structured proposals, deterministic validation, a tracked verification ledger, and human or independent review for ambiguous cases. The central design rule is operational rather than aspirational: AI proposes; evidence decides.

As of 31 August 2026, the canonical snapshot contains 993 records at 423 universities in 20 countries, organized into 17 broad fields and five appointment tracks, spanning recorded PhD years from 1962 to 2026. Runtime aggregation exposes institutional and disciplinary patterns, while Git history records additions, promotions or track corrections, moves, removals, and deduplication. The system is not a complete census and does not provide controlled accuracy or cost estimates. Its contribution is twofold: a public resource for a community that previously had none, and a concrete engineering pattern showing that a public directory is a continuously revalidated collection of claims about an external world, not merely a database snapshot.

1 Introduction

Vietnamese-diaspora—people who are either Vietnamese or have Vietnamese heritage—faculty are spread across hundreds of universities in dozens of countries outside Vietnam. Despite the growth of this population, there is no public, searchable record of it. A prospective graduate student looking for a Vietnamese advisor in their field, a journalist profiling the diaspora, or a scholar looking for compatriots at nearby institutions currently has no better option than ad hoc Google search, one name at a time. Communities of comparable size often sustain an informal directory through an alumni association or professional society, where expatriate researcher networks support mentorship, collaboration, and technology diffusion back toward their country of origin [11, 8].

VietProfs was built to be that resource: a public, searchable directory of Vietnamese and Vietnamese-diaspora faculty, useful only if it stays current as people are promoted, move, or retire. By leveraging AI LLM agents to find and propose updates, and verify them against the public web, the open-source system maintains itself with minimal human intervention and therefore reduces the risk of being out of date, neglected or abandoned.

1.1 What VietProfs Is: Motivation, Goals, and Design

Before this project, a Vietnamese-heritage scholar could be found only one search at a time, on their own university's page, with no way to see them alongside their peers. VietProfs closes that gap and helps connect people across institutions and disciplines. For example, it can help a graduate student find an advisor who shares their background or research interest. It lets a faculty member discover Vietnamese-diaspora colleagues at a new institution or nearby ones. It gives conference organizers, professional societies, and student or alumni associations a way to identify speakers, panelists, and committee members. It gives journalists and university or cultural offices a single place to start when documenting the diaspora's reach. And, independent of any specific task, it lets the community—and outsiders—see its own scale and range across institutions, countries, fields, and generations, from Xuong Nguyen-Huu, who earned a 1962 PhD at UC Berkeley and now holds an appointment at UC San Diego, to the newest hires in the current snapshot (Section 2).

Inclusion is governed by a single, name-independent rule: a current academic appointment at a university or eligible public/nonprofit scholarly research institute outside Vietnam, in one of six accepted tracks—Tenure-line, Teaching, Research, Clinical, Academic staff, or Emeritus. Each track is defined precisely enough that a stable, full-time, faculty-level role is included, while a temporary, visiting, postdoctoral, or adjunct one is not (Section 3). Every included person is also classified into one of 17 broad fields of study, broadly classified by the U.S. National Center for Education Statistics, National Science Foundation, and National Institutes of Health. The goal is for the directory's field distribution to read against familiar national categories (e.g., Computing, Mathematics, Health, Social and Behavioral Sciences) rather than an invented scheme (Section 4).

As of 31 August 2026, the canonical snapshot holds 993 records at 423 universities in 20 countries, across all 17 fields and all five tracks (Section 8). Each person's displaying card is designed around what a visitor actually needs, such as a portrait, Vietnamese-diacritic name, current university, department, rank, and location, education history, curated honors, and short research-area tags. Curated honors are limited to real field-wide distinctions such as national-academy election or a major career award. Short research-area tags support topic-level search, and a visible last-verified date lets a reader judge how current a given fact is (Section 5). The site itself supports free-text search that can be narrowed to a single attribute (including Name, Institution, Department, Rank, Research area, Honors, or PhD institution), combinable location/field/track/institution-type filters, shareable filtered URLs, institution and PhD-institution leaderboards, a state-by-state map of U.S. records, and a public form for anyone to submit a new person or a correction (Section 7).

1.2 The Engineering Challenge: Keeping VietProfs Accurate

Building a first version of this directory would not, by itself, have been hard. Keeping it accurate is a much harder engineering problem, and it is the technical contribution of this paper. Structured public datasets become stale for a simple reason: the world they describe changes while the dataset waits for an update. A person becomes a new faculty, an existing faculty member gets promoted, changes universities, becomes emeritus, or leaves academia. A URL or image disappears in a site redesign. A newly eligible person never appears in the search results that originally seeded the directory. Maintenance is a reconciliation problem between those claims and a distributed, changing web (Section 10), whose pages are known to disappear or change at a substantial rate over just a few years

[3]. Manual human maintenance would be very difficult and do not scale, because it would require a human to check hundreds of records against the web and even try to do things beyond the capability of Google search, such as finding a newly eligible person who has not yet been indexed or surfaced by any search engine.

VietProfs (https://vietprofs.roars.dev) addresses this with an automatic maintenance system that employs continuous loops of expansion and validation—*all using AI agents*. Expansion searches a bounded university/field/country slice for people not yet in the roster. Revalidation periodically re-checks every existing record's profile link, appointment, rank, education, and honors against the current web (Section 14). The system's loops run continuously, almost every day, to keep the roster current. VietProfs is an AI sytem itself, and it would not exist without the LLMs that power it.

That said, AI are not a magic wand, and human contribution is still essential. One of the surprising lessons of this project is that finding *new* information is very difficult and time-consuming, even for state-of-the-art LLMs, but checking a claim against the web is much easier. For example, given the relatively complete roster of faculty already in VietProfs, it is difficulty for LLMs to scout the web for new, previously unlisted people (and in fact, the agents often complained that scouting is a wasteful task with little yields after spending hours of researching and only to find candidates who already exists in the database). Thus, VietProfs also gets frequent contributions from human users who submit new people or corrections through the public form (Section 7.3), and the LLMs simply need to check the submitted information against the web and verify it, which is much easier and more reliable. In addition, LLMs experience hallucination, e.g., putting pictures of the wrong person happens quite often, so human spot-checking is still needed (Section 11).

This paper has two aims, addressed in two parts. Part I is about VietProfs itself: who it is for, who is eligible to appear, what fields and kinds of information it covers and why, how it is used, and what it currently shows. This is the paper's main subject. It is written for a reader who cares first about the directory, and only secondarily about how it is built or maintained. Part II turns to engineering. Keeping several hundred records accurate against a constantly changing web required a nontrivial AI-assisted maintenance system, and we report that system as a bounded technical contribution in its own right: a data model and operational architecture (Section 6 onward); a workflow separating candidate discovery from evidence-backed verification (Sections 11–13); an empirical characterization of expansion and revalidation from implementation artifacts and public Git history (Section 14); the derived-knowledge maintenance problem created by runtime aggregation (Section 8); and the test suite that lets an unattended controller push directly to production (Section 15).

VIETPROFS IS LIVE AND SEARCHABLE AT
https://vietprofs.roars.dev

Part I
VietProfs: Purpose, People, and Content

2 Goals and Intended Uses

VietProfs exists to answer a concrete, recurring need: given a Vietnamese-heritage scholar somewhere in the world, or an interest in the diaspora's presence at a given university or in a given field, is there anywhere to look? Before this project, the honest answer was no. The site is designed around a small number of intended uses, each already reflected in how the interface is built (Section 7).

- **Finding a mentor or advisor.** A prospective graduate student who would like a Vietnamese-speaking advisor, or who simply values shared cultural background in a mentoring relationship, can search by field and location instead of guessing names from department web pages one at a time—a search for engineering, for instance, surfaces

Nam-Trung Nguyen at Griffith University and Tho Le-Ngoc at McGill University alongside dozens of others.

- **Building community across institutions.** A professor moving to a new country, or simply curious whether there are other Vietnamese-diaspora faculty nearby, can search by university, city, state, or country and find colleagues they would otherwise never learn about; a newcomer to the National University of Singapore would find Nhan Phan-Thien, Hai Minh Duong, and Han Vinh Huynh already listed there, and a newcomer to Paris would find Duong Hieu Phan at Télécom Paris and Phu Nguyen-Van at Université Paris Nanterre.
- **Assembling speakers, panels, and committees.** Conference organizers, professional societies, and Vietnamese student and alumni associations can use the field and honors filters to identify potential keynote speakers, panelists, or program-committee members with relevant expertise—a panel on Vietnam War history could draw on Lien-Hang T. Nguyen (Columbia) and Nu-Anh Tran (University of Connecticut).
- **Documenting representation.** Journalists, university offices, and cultural or diplomatic organizations that want to describe the reach of Vietnamese scholarship abroad have, for the first time, a single place to start rather than an ad hoc collection of anecdotes—from Lan Cao's professorship of law at Chapman University to Nam-Trung Nguyen's Australian Laureate Fellowship at Griffith University.
- **A record the community can see itself in.** Independent of any specific task, the directory is a form of collective visibility: it lets the diaspora, and outsiders, see the range of institutions, countries, fields, and generations—the earliest recorded PhD year in the current snapshot, 1962, belongs to Xuong Nguyen-Huu, who now holds an appointment at UC San Diego—that Vietnamese-heritage academics have reached.
- **A starting point for description** The aggregated view (Section 8) lets a visitor ask simple descriptive questions—which fields, which countries, which institutions are represented in the maintained roster, such as Monash University's 13 faculty or the 60 records currently located in Australia—while Section 9 explains why that view should not be read as a population survey.

VietProfs is deliberately not several other things it might be mistaken for. It is not, and never will be, a ranking system. It is also not a census of the Vietnamese academic diaspora: Section 9 is explicit that coverage depends on search effort and discoverability, so the roster documents who has been found and verified, not the full population.

3 Who Is Included: Eligibility

A directory is only as useful as its inclusion rule is clear, and only as trustworthy as that rule is applied consistently. This section states, in plain terms, who is meant to appear in VietProfs and why; Section 6 gives the same rule again as it is literally encoded in the data-entry and validation logic that enforces it.

3.1 The core requirement

A person is included only if reliable evidence—preferably an official institutional page—supports two things at once: (1) a current academic appointment at a university or eligible public/nonprofit scholarly research institute anywhere outside Vietnam (with a narrow exception for Emeritus, discussed below), and (2) that appointment falling into one of six accepted tracks, summarized in Table 1.

Outside Vietnam is a scope decision, not a value judgment about scholars based in Vietnam: the directory is specifically about the diaspora, a population scattered across hundreds of institutions in dozens of countries with no shared index, which is the gap this project set out to fill. Besides, it also does not make too much sense to have a directory of Vietnamese professors *in* Vietnam, because most likely every professor in Vietnam is Vietnamese, and the official university pages already provide a complete index of them.

Each track has a real counterpart already in the roster: Nam-Trung Nguyen holds a Tenure-line professorship in engineering at Griffith University; Thanh Tran is a Professor in the Practice of electrical and computer engineering at Rice University under the Teaching track; Khiem Pham-Nguyen holds a Clinical Assistant Professorship in orthodontics at Boston University; and Xuong Nguyen-Huu, whose 1962 PhD from UC Berkeley predates every

Table 1: Accepted appointment tracks and their intent (`ROSTER_MAINTENANCE.md`).

Track	Who it covers
Tenure-line	Tenure-track or already-tenured faculty (Assistant, Associate, or full Professor).
Teaching	Stable, full-time, continuing non-tenure-track teaching faculty, including Professor of Practice and equivalent titles, confirmed by the institution's own language (e.g., "full-time," "continuing appointment," a named promotion ladder) rather than inferred from the title alone.
Research	A stable, faculty-level or faculty-equivalent appointment at a university or eligible public/nonprofit scholarly research institute—not a postdoctoral, visiting, grant-limited, or otherwise temporary research role.
Clinical	A stable clinical-faculty appointment such as Clinical Professor or a documented clinical-faculty ladder—not an adjunct or temporary clinical-teaching role.
Academic staff	A university librarian or archivist with documented faculty status or a senior, permanent academic appointment—not an ordinary staff or temporary role.
Emeritus	A formally conferred emeritus/emerita title following a tenure-line career, evidenced by an active emeritus listing or a documented conferral—not plain retirement, resignation, or an in-memoriam page.

other recorded degree in the snapshot, holds an Emeritus appointment at UC San Diego.

3.2 Who is not included, and why

The tracks deliberately exclude several kinds of appointment that a casual search would otherwise surface: adjunct, visiting, postdoctoral, affiliate/courtesy, graduate teaching-assistant, industry-only, and other term-limited or part-time positions. Corporate research labs, including Microsoft Research, remain outside scope; eligible non-university employers are limited to public research bodies and independent nonprofit scholarly research institutes, whose entries are explicitly labeled by institution type. This is not a judgment about the people who hold these roles—many are excellent scholars, and several will later hold an included appointment—but a decision about what a stable directory entry can promise. A visiting or postdoctoral appointment typically lasts one to three years and is, by design, temporary; including it would either commit the directory to constant churn or invite stale entries for people who have long since moved on. A plain `Instructor` title is treated case by case rather than included from the title alone, because the same word covers everything from a permanent teaching-track position to a single-semester contract depending on the institution.

Two titles deserve special mention because they are easy to misjudge from the outside. `Research Assistant Professor` is included only when the institution treats it as a genuine research-faculty rank, evidenced by, for example, a departmental faculty directory listing or an established promotion ladder—not merely because the word "Professor" appears in it, since several institutions use exactly this title for what is functionally a senior postdoctoral position. Professor of Practice and its equivalents (Associate/Assistant Professor of Practice) are included under the Teaching track once their stability is confirmed, with the institution's own practice title preserved in the record rather than flattened into a generic rank.

3.3 Identity is not inferred from a name

VietProfs uses Vietnamese given names and common surnames—`Nguyen`, `Tran`, `Le`, `Pham`, `Vo`, `Vu`, `Bui`, `Do`, `Phan`, `Lai`, among others—as a starting point for finding candidates, the same way a person doing this search by hand would. But a name is a lead, never a verdict: once a candidate is otherwise eligible under Table 1, no separate documentary proof of Vietnamese ethnicity or national origin is required or requested. This is a deliberate choice on both ethical and practical grounds. Ethically, demanding proof of heritage from a public figure in order to list them in a public directory would be invasive in ways this project has no interest in enforcing. Practically, a surname is neither sufficient evidence of identity—plenty of people who share a common Vietnamese surname are not Vietnamese, and some Vietnamese-heritage scholars do not have one, for example after a name change through marriage or adoption—nor a reliable way to tell two different people apart. The general unreliability of inferring ethnicity or origin from a name alone is well documented in the name-based ethnicity classification literature, which treats a surname at best as a probabilistic signal requiring corroboration, not a determination [10, 1]; VietProfs's policy of treating a name only as a discovery lead is a conservative application of that same caution. This is also why the directory deduplicates by verified person rather than by name string (Section 6). A person who does not wish to appear, or who has been mis-identified, can request removal or correction through the same public channel used to propose additions (Section 7.3).

4 Fields of Study Covered

To make roughly a thousand records browsable by discipline, every person in VietProfs is classified into one of 17 broad fields (Table 2). Rather than inventing an idiosyncratic taxonomy, these 17 groups are a coarsened version of field categories already used in U.S. federal statistical practice, so that a visitor familiar with how universities or funding agencies group disciplines will find the groupings unsurprising, and so that VietProfs's field distribution can be discussed in the same vocabulary as national statistics on doctoral education. Two federal classification systems anchor this correspondence:

- The National Center for Education Statistics' *Classification of Instructional Programs* (CIP), the standard scheme U.S. institutions use to report degree programs to the federal government, organizes fields into roughly 40 two-digit "families" (e.g., `11` Computer and Information Sciences, `14` Engineering, `26` Biological and Biomedical Sciences, `51` Health Professions and Related Clinical Sciences) [12].
- The National Science Foundation's Survey of Earned Doctorates, run by its National Center for Science and Engineering Statistics, groups doctorates into a smaller set of broad fields—life sciences, physical sciences and earth sciences, mathematics and computer sciences, psychology and social sciences, engineering, education, and humanities and arts, among others—for its annual reporting on U.S. doctorate recipients [13].

Table 2 shows, for each VietProfs field, the nearest corresponding CIP family. The mapping is a coarsening for browsability, not a formal crosswalk: several VietProfs fields merge more than one CIP family (for example, Humanities merges English/literature, foreign languages and linguistics, history, and philosophy), and a few disciplines sit at a boundary the source taxonomies themselves treat inconsistently—economics, for instance, is filed under Social Sciences in CIP (45.06) even though it is more often taught and hired alongside business, which is why VietProfs groups it with Business rather than with Social & Behavioral Sciences.

Health Sciences is large enough, and clinically prominent enough in the roster specifically—182 honors-bearing records, 96 Clinical-track appointments, and several of the roster's most senior appointments are in medicine—that VietProfs further splits it into derived subfields (Clinical Medicine, Public Health, Nursing, Pharmacy, Dentistry, Biomedical Research, and Medical Education; Section 6). Table 3 aligns those clinical subfields with their closest counterpart among the National Institutes of Health's 27 Institutes and Centers, each of which funds and organizes research around a specific organ system, disease category, or health discipline [14]. The alignment is again illustrative rather than exact: several NIH Institutes (for example, the National Cancer Institute or the National Heart, Lung, and Blood Institute) fund research across what VietProfs simply calls Clinical Medicine, and population/prevention health in the U.S. is coordinated at least as much through the Centers for Disease Control and Prevention as through NIH; the table cites the clearest correspondences rather than forcing every subfield onto a single Institute.

This classification exercise is a browsing aid, not a scientific claim. VietProfs's own field-mapping code (`fieldOf`, Section 6) works from department names and an explicit override table, and a handful of institution-specific department names will always sit awkwardly between two standard categories; the maintenance guide's response to that ambiguity is to ask rather than guess (Section 6).

Table 2: VietProfs's 17 broad fields and their nearest correspondence in the NCES CIP family structure [12]. A VietProfs field may draw on more than one CIP family; the table lists the primary one(s).

VietProfs field	Nearest CIP family/families
Computer & Information Sciences	11 Computer and Information Sciences and Support Services
Engineering	14 Engineering; 15 Engineering Technologies
Mathematics	27 Mathematics and Statistics (pure/applied mathematics)
Statistics & Data Science	27 Mathematics and Statistics (statistics/data science)
Physics & Astronomy	40 Physical Sciences
Chemistry	40 Physical Sciences
Biological & Biomedical Sciences	26 Biological and Biomedical Sciences
Earth & Environmental Sciences	40 Physical Sciences (earth/atmospheric/ocean sciences); 03 Natural Resources and Conservation
Agricultural & Natural Resource Sciences	01 Agriculture, Agriculture Operations, and Related Sciences; 03 Natural Resources and Conservation
Health Sciences	51 Health Professions and Related Clinical Sciences (see Table 3 for subfields)
Business & Economics	52 Business, Management, Marketing; 45.06 Economics
Social & Behavioral Sciences	45 Social Sciences; 42 Psychology
Education	13 Education
Humanities	23 English Language and Literature; 16 Foreign Languages and Linguistics; 54 History; 38 Philosophy and Religious Studies
Law & Public Affairs	22 Legal Professions and Studies; 44 Public Administration and Social Service Professions
Arts & Design	50 Visual and Performing Arts; 04 Architecture and Related Services
Others	30 Multi/Interdisciplinary Studies; institution-specific fields not yet mapped

5 What Information Is Shown, and Why

Every field stored for a person (Table 4, Section 6) earns its place because of what it lets a visitor do, not because the underlying research happened to surface it. This section explains the reasoning behind the fields a reader will actually see on a card (Figure 1).

Portrait. A small photo turns a row of text into a person. For a directory whose purpose is partly about recognition and connection—a student wondering "is that the same Professor Nguyen I met at a conference?", a colleague scanning a list of new faculty at a nearby university—a face is often the fastest way to confirm identity, especially given how common several Vietnamese surnames are. Portraits are stored locally with their source URL recorded (`portraitSource`), so provenance is preserved even though the image itself is not a link that can go stale.

Vietnamese-diacritic name. Many official English-language university pages render a person's name without diacritics, or in Western given-name/surname order. VietProfs stores the diacritic form (`vietnameseName`) alongside the display name specifically so the directory reflects how a person's name is actually written in Vietnamese, which is both a form of respect and, practically, another way for a Vietnamese-speaking visitor to recognize a name they already know.

University, department, rank, and location. This is the directory's most basic utility: knowing that someone exists is not the same as knowing where to find them. Location fields also power the browsing filters and the state/country leaderboards in Section 7.

Education (undergraduate, master's, PhD, postdoctoral, and other professional degrees). Recording where and when someone trained does two things. Practically, it helps a student identify a potential mentor who trained in a similar path or at a familiar institution. Collectively, it is also a quiet piece of evidence against the assumption that Vietnamese-diaspora academic presence is recent: the earliest recorded PhD year in the current snapshot is 1962, more than six decades before this paper, spanning generations of scholars rather than a single wave.

Honors. Recognitions are shown to celebrate documented excellence and to make role models visible to students

Table 3: VietProfs Health Sciences subfields and their closest NIH Institute/Center counterpart, from the official list of NIH Institutes, Centers, and Offices [14], or the nearest CIP subfamily where no single Institute corresponds [12].

VietProfs subfield	Closest federal classification
Nursing	National Institute of Nursing Research (NINR)
Dentistry	National Institute of Dental and Craniofacial Research (NIDCR)
Biomedical Research	National Institute of General Medical Sciences (NIGMS); National Center for Advancing Translational Sciences (NCATS)
Clinical Medicine	Collectively, the disease- and organ-focused Institutes (e.g., National Cancer Institute; National Heart, Lung, and Blood Institute; National Institute of Diabetes and Digestive and Kidney Diseases)
Public Health	National Institute of Environmental Health Sciences (NIEHS) within NIH; population and prevention health more broadly is coordinated by the CDC, outside NIH
Pharmacy	No single dedicated NIH Institute; corresponds to CIP 51.20 (Pharmacy, Pharmaceutical Sciences, and Administration)
Medical Education	CIP 51.12 (Medicine); not a distinct NIH Institute

and peers who might not otherwise learn about them—seeing that a member of the community was elected to the National Academy of Medicine or won a Sloan Research Fellowship is itself meaningful to a community whose achievements are rarely aggregated anywhere. Because a personal CV can list dozens of items of wildly different weight, honors are curated rather than imported wholesale; Section 6.1 explains exactly what qualifies and what does not.

Research areas. Short topic tags let a visitor searching for, say, "machine translation" or "coral reef ecology" find the right person even when a department name alone ("Computer Science," "Biology") would not distinguish them.

Scholar and personal/lab links. These give a visitor somewhere to go next—publications, a lab website, contact information—once the directory has done its job of helping them find the right person.

`lastUpdatedAt`. Because the people and pages a directory describes keep changing (Section 10), every card shows when its content was last confirmed to be current, so a visitor can judge how much to trust a given detail rather than assuming a static database is automatically up to date.

Two things are deliberately *not* shown. VietProfs does not display any inferred or asserted ethnicity, birthplace, or immigration information beyond what a person's own official professional page already states; the directory is about a current academic appointment, not a biography. And it does not display any ranking, score, or comparative judgment across people or institutions—the leaderboards in Section 7 are counts, not rankings of quality.

6 VietProfs and the Data Model

The public source of truth is `public/data.json`; the browser loads it and performs search, filtering, sorting, and aggregation. Table 4 lists the record schema implemented by the `RosterEntry` TypeScript interface in `src/data.ts`. A record contains an immutable `vp-` identifier followed by digits, a name, current profile URL, employing institution and optional non-university `institutionType`, city, state or province when applicable, country, department, research areas, rank and appointment track, optional education across five credential types (undergraduate, master's, PhD, postdoctoral, and professional degrees such as MD), optional honors with source citations, optional Scholar and personal-site URLs, portrait provenance, and `lastUpdatedAt`. The separate `maintenance/verification.json` ledger records the time of a completed full review, keyed by canonical name. It is not part of the public JSON payload, so maintenance state travels with the repository across machines without inflating what a browser downloads.

The inclusion rule is a classification policy, restated here as it is literally implemented after being introduced for a general reader in Section 3. A person must have a current primary appointment at a university or eligible public/nonprofit scholarly research institute outside Vietnam, except for qualifying emeritus appointments, and must

Table 4: Public roster schema (`RosterEntry`, `src/data.ts`). Required fields are used by every record; optional fields are present when a source explicitly documents them.

Field group	Fields
Required	`id`, `name`, `university`, `department`, `rank`, `track`, `profileUrl`, `lastUpdatedAt`, `researchAreas[]`
Location	`city`, `state`, `country`
Identity	`vietnameseName` (diacritic form)
Institution	`institutionType` (required for eligible non-university institutes)
Education	`undergradInstitution/Year/Major`, `msInstitution/Year/Major`, `phdInstitution/Year/Major`, `postdocInstitution/Year`, `mdInstitution/Year`, `otherDegrees[]` (`degree`, `institution`, `year`, `major`, `source`)
Honors	`honors[]`: `name`, `organization`, `category`, `year`, `source`
Links	`websiteUrl` (personal/lab), `scholarUrl` (Google Scholar)
Portrait	`portrait` (local `.webp` path), `portraitSource` (provenance URL)

fit one of six tracks: Tenure-line, Teaching, Research, Clinical, Academic staff, or Emeritus. Adjunct, visiting, postdoctoral, affiliate/courtesy, temporary, part-time, industry-only, and corporate-research positions are excluded. Teaching includes stable full-time non-tenure-track appointments such as Professor of Practice; Research and Clinical require evidence that the appointment is faculty-level and stable; Emeritus requires a formally conferred title following a tenure-line career. The published rank is normalized for the public vocabulary where appropriate, while institution-specific titles are preserved when they clarify a non-tenure appointment. [CODE: `ROSTER_MAINTENANCE.md`; `src/data.ts`]

Vietnamese identity is not inferred from a surname. Vietnamese names and common surnames are used as search signals, but eligibility depends on the appointment and the project's inclusion policy. This distinction matters for both ethics and correctness: `Nguyen` is neither sufficient evidence of identity nor a reliable duplicate key. Names are deduplicated by person, with a university suffix used only when genuinely identical names require disambiguation.

Departments are mapped to 17 broad fields by ordered rules and institution-specific overrides. Health Sciences additionally has derived subfields such as Clinical Medicine, Public Health, Nursing, Pharmacy, Dentistry, and Biomedical Research. The mapper is deliberately explicit because strings such as "Information Studies" or "Mathematics and Statistics" can mean different things in different institutional contexts. The tests assert both the taxonomy and representative overrides. [CODE: `src/data.ts`; `test/data.test.ts`]

6.1 Honors and awards eligibility

The `honors` field is curated for substantial distinctions, not every item listed on a CV or personal homepage. Table 5 lists the five accepted categories; both the controller's proposal validator and `scripts/validate-data.ts` enforce that every stored honor declares one of them, together with a name, organization, HTTPS source, and—when known—a year between 1900 and the current year. An award created and administered by a single university is presumed local and ineligible unless independent field-wide standing is documented; a large-sounding title or cash prize alone is not evidence of that standing.

Concretely, an included honor looks like election to the National Academy of Engineering, National Academy of Medicine, or an equivalent learned academy; Fellow status from IEEE, ACM, AAAS, or a comparable major disciplinary society; a nationally competitive early-career award such as an NSF CAREER award, a Sloan Research Fellowship, a Simons Investigator award, or PECASE; a major field-wide prize, medal, or test-of-time award; or a named endowed chair or distinguished professorship. An excluded honor looks like a best-paper award from a single workshop, a departmental teaching or service award, an "outstanding reviewer" certificate, an internal university seed grant, a list of invited talks, or a large-sounding institution-run prize whose independent standing outside that one university cannot be confirmed. The line is deliberately conservative: a shorter, verifiable list of real distinctions

Table 5: Accepted honor categories (ROSTER_MAINTENANCE.md, enforced by proposalValidationError and validate-data.ts).

Category	Qualifying pattern
academy	Election to a recognized national academy or equivalent learned academy.
fellow	Fellow or honorary-member status conferred by a major disciplinary society.
career_award	A nationally/internationally competitive career or early-career award (e.g., NSF CAREER, Sloan Research Fellowship, Simons Investigator, PECASE).
major_award	A field-wide medal, prize, book award, or lifetime/impact/test-of-time distinction.
distinguished_professorship	A named endowed chair, distinguished professorship, or comparable research chair.

is more useful to a reader than a long list padded with routine recognitions that are hard to compare across people.

7 User Interface and Interaction Model

Because the roster is only useful if the population it describes can actually find themselves and each other in it, the client-side search and browsing interface is a first-class part of the system, not a thin wrapper over the JSON file. Figure 1 shows the landing view: a free-text search box with field-specific keyword scopes, four combinable filters (location, field, appointment track, and institution type), a row of example queries generated from the current roster, a result count, and a card per matching person showing name, Vietnamese-diacritic form, current rank/department/institution/location, education, honors, research tags, and a `lastUpdatedAt` badge so a visitor can see how fresh a given record is.

7.1 Scoped search and diacritic normalization

The search box matches names, institutions and institution types, departments, ranks, research areas, honors, and PhD institutions by default, with matching performed on a diacritic-stripped, lowercased index (`buildSearchIndex`/`normalizeSearchText` in `src/data.ts`) so that typing `Nguyen` finds `Nguyễn`. Keyword prefixes can narrow the same text to Name, Rank, Research area, Honors, University, Institution, Department, or PhD institution. Location, field, track, and institution type are exposed as dedicated combinable filters. Small finite scopes expose their available values immediately as suggestions. The active search text, selected scope, and filters are mirrored into the URL query string, so a filtered view—for example, all Clinical-track faculty in Health Sciences at a named university—is directly shareable as a link rather than requiring the recipient to reproduce the same clicks.

7.2 Leaderboards, geography, and the derived-facts view

Below the roster list, the site computes "Top Faculty Hubs" and "Top PhD Alma Maters" leaderboards for the currently selected subset, each entry itself a clickable scoped filter (the search text and scope are carried separately) so browsing and querying are the same action. Selecting a U.S. location, or the sentinel "*Show me something interesting*" option in the field filter, additionally renders a state-by-state choropleth grid (`src/state-grid.ts`) where tile shade encodes record count and a click filters to that state; Figure 4 shows this view. This is discussed further as the site's third function, runtime aggregation, in Section 8.

7.3 Human contribution channel

Automated maintenance (Section 11) is the primary way the roster grows, but a visitor can also propose a new person or a correction directly, without editing the repository, via `submit.html` (Figure 3). The form requires only

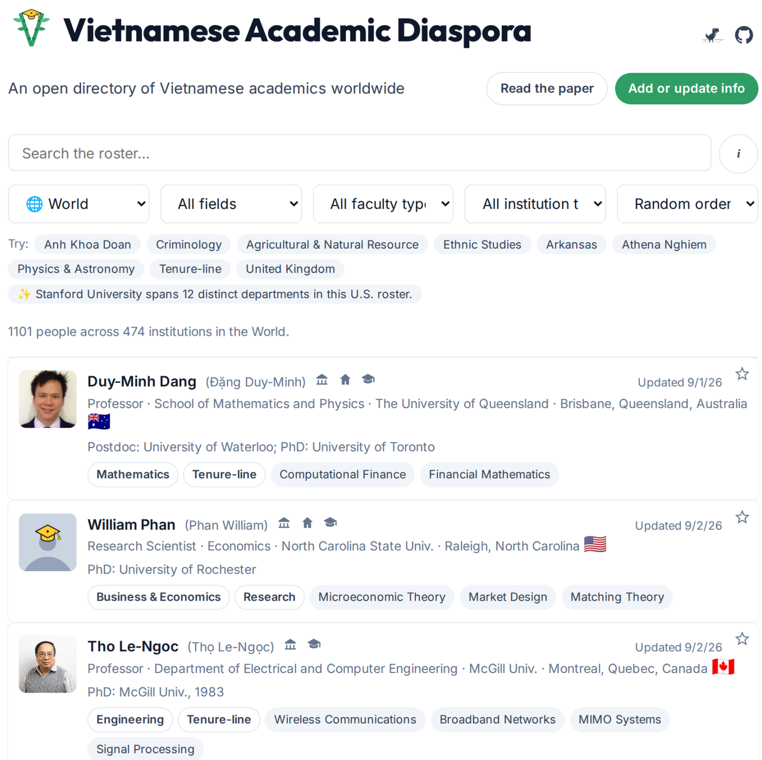

Figure 1: The VietProfs landing view: search, combinable location/field/track filters, example query chips drawn from the live roster, and per-person cards with rank, department, university, location, education, honors, and research tags.

a full name and at least one verification link (a university profile, personal site, LinkedIn, or similar—the visitor need not already have the official university page in hand); typing a name that already resolves against the loaded roster (`src/submit.ts`) surfaces a suggestion and pre-fills the existing record so the visitor is editing rather than duplicating. Each generated profile page also links directly to this pre-filled correction form using the immutable ID. A correction email carries that ID, the permanent profile URL, current and proposed names, and a field-by-field delta, so a name correction remains unambiguous for a maintainer. Every other field—Vietnamese name, personal/lab site, official university profile page, Scholar profile, department, location, education, honors—is optional, and a single free-text box doubles as either a bulk paste of names/links for new candidates or freeform notes and evidence for a correction. Submissions are routed to a maintainer inbox and reviewed before they can change canonical data; nothing typed into this form writes to `public/data.json` directly. This keeps the boundary from Section 13 intact for human-sourced leads as well as agent-sourced ones: a submission is evidence and a lead, not an approved change.

8 What the Directory Reveals

The site has a third function beyond building and maintaining knowledge: learning from the current roster. Primary knowledge is the individual record. Aggregation produces derived knowledge such as institutional concentration, field balance, geography, track distribution, awards, and doctoral institution patterns.

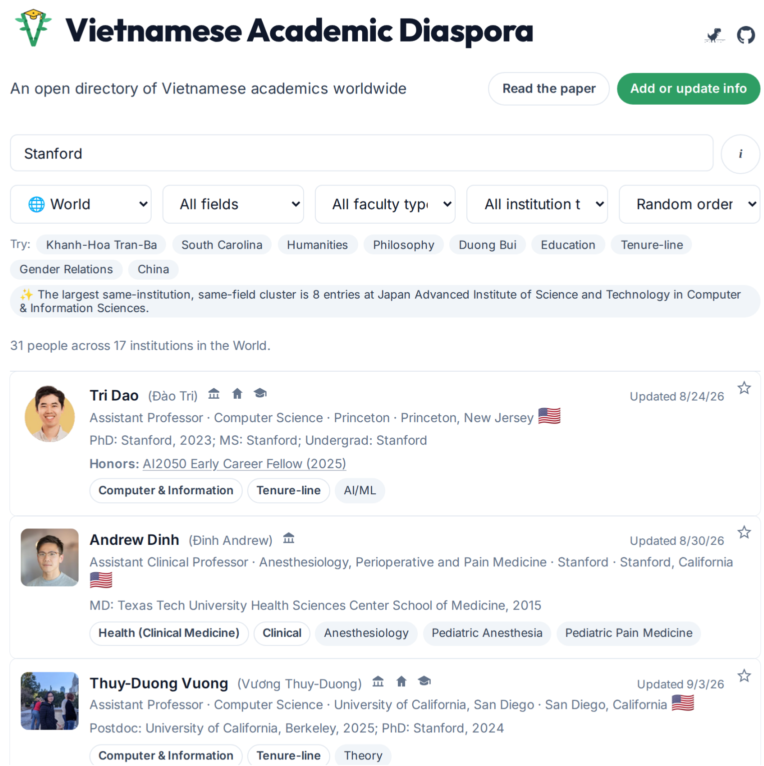

Figure 2: A university-scoped search (the scope selector set to University) narrowing the full roster to the 14 Stanford entries, each with rank, department, location, education, research tags, and a last-verified badge.

8.1 Runtime observations

The "Show me something interesting" view (Figure 4) is computed at runtime from the current JSON snapshot by `buildFunFacts` and a family of `build*Observations` functions in `src/data.ts`, rendered by `src/main.ts`. It reports U.S. and international counts, state/country/institution groups, same-institution same-field clusters, cross-department breadth per institution, field comparisons between the U.S. and international subsets, award summaries, PhD cohorts by decade, and the same university/PhD-institution leaderboards described in Section 7, here rendered for the full worldwide roster rather than a filtered subset. It stores no hand-written facts. Tests require multi-record, roster-only observations and reject unsupported surname, prestige, demographic, causal, or population claims; the maintenance guide is explicit that a current-roster concentration must not be described as growth, an emerging region, or a historical trend without a versioned historical dataset to support it. The result is therefore a descriptive view, not an estimate of the entire diaspora. [CODE: `src/data.ts`; `src/main.ts`; `test/data.test.ts`]

8.2 Snapshot observations

Table 6 summarizes the current snapshot. Within it, Stanford University and UC San Diego each have 14 records, followed by Monash University with 13; UC Irvine, UCLA, and the University of Florida each have 11, while NUS, Texas Tech, the University of Melbourne, the University of Toronto, and the University of Wisconsin–Madison each have 10. California and Texas contain 218 of 659 U.S. records (33%). The international subset contains 334 records across 19 non-U.S. country values, led by Australia (64), the U.K. (54), Canada (47), France (36), Japan (33), Singapore (25), and Taiwan (21). The four largest broad fields are Health Sciences (203), Computer and Information Sciences (148), Business and Economics (143), and Engineering (139); the U.S. and international subsets differ sharply in their leading field, at 27% Health Sciences and 22% Computer and Information Sciences respectively.

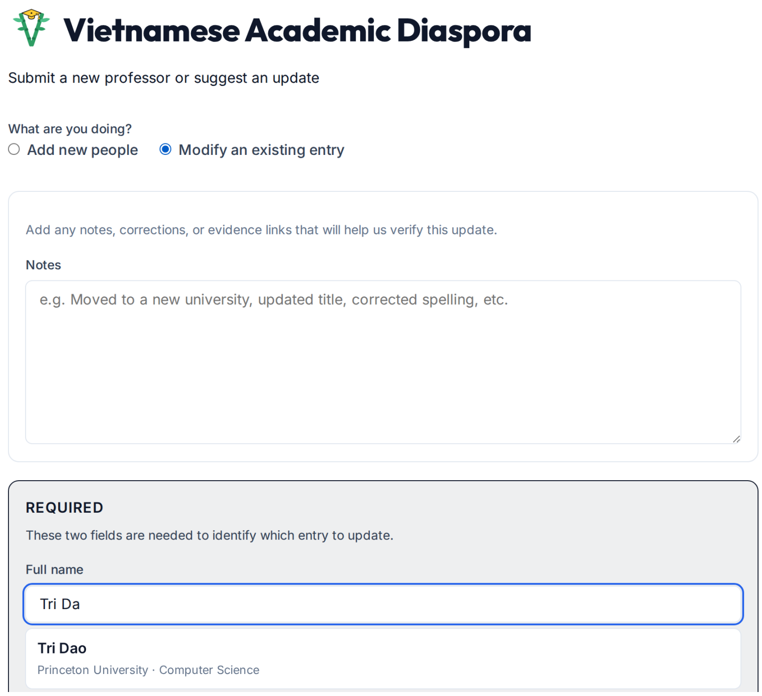

Figure 3: The public submission form (`submit.html`). Only a name and one verification link are required; a name that matches an existing record is offered back to the visitor as a correction rather than a duplicate.

Table 6: Current VietProfs snapshot and evidence-field coverage.

Records	993	Universities	423
Countries	20	U.S./international	659/334
Broad fields	17	Accepted tracks	5
Profile URLs	993	Scholar URLs	219
PhD institutions	751	PhD years	549
Honors-bearing records	182	PhD-year range	1962–2026

Stanford has 13 health-science records; the Japan Advanced Institute of Science and Technology has eight computing records; and the University of Minnesota and the University of Technology Sydney have seven each, in health sciences and engineering respectively. Monash and Stanford each span 12 distinct stored departments, and UC Irvine spans 11, showing that some institutional clusters are cross-disciplinary rather than concentrated in one department. These are useful because they reveal institutional and disciplinary structure without claiming prestige, population size, or causation.

8.3 Derived knowledge also becomes stale

If a person moves, an institutional count changes. If a track is corrected, the track distribution changes. If a record is removed, a field comparison or same-institution cluster may no longer hold. The current implementation recomputes the visible observations from the live roster at render time, so the runtime view does not preserve stale fact text. It does not maintain a separate dependency graph for persisted derived facts because those facts are not persisted. A future system that stores notable findings would need dependencies, recomputation, and invalidation policies just as primary records do.

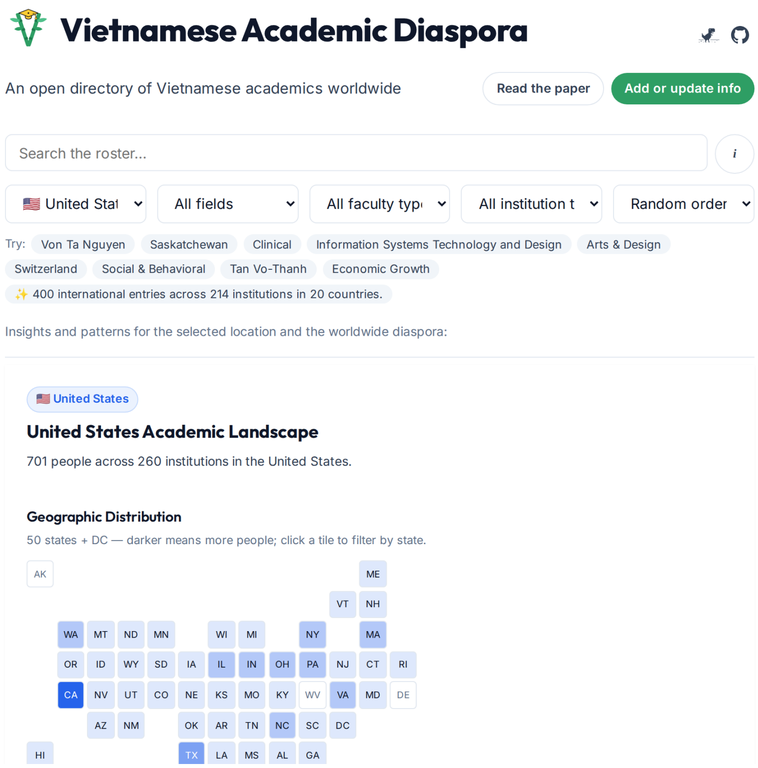

Figure 4: The runtime-computed “Show me something interesting” view for the United States subset, including the state-level choropleth grid (`src/state-grid.ts`); darker tiles indicate more records, and clicking a tile filters the roster to that state.

9 Limitations and Ethics

VietProfs is not a census. Coverage is affected by search effort, English-language discoverability, university web quality, public-profile availability, initial field emphasis, international title conventions, and the differing maturity of collection passes. The 659/334 U.S./international split must not be interpreted as a population ratio. Nor should current concentrations be interpreted as prestige, demographic size, causation, migration history, or research quality.

The directory handles sensitive identity-adjacent information. It should not infer ethnicity or national origin from names, and it should publish only public professional information necessary for the directory’s purpose. Portraits, honors, and education need source and scope rules. Public submission and correction mechanisms (Section 7.3) help people contest errors, but a future release should expose clear removal/contact policy, retention expectations, and an audit trail for disputed changes.

Part II

Building and Maintaining VietProfs: A Technical Report

Part I described what VietProfs is, who it includes, what it covers, and why. The remainder of this paper turns to engineering: how a small team keeps hundreds of records accurate against a constantly changing web, using AI-assisted automation, deterministic validation, and human oversight, without that automation becoming an unaccountable source of truth in its own right.

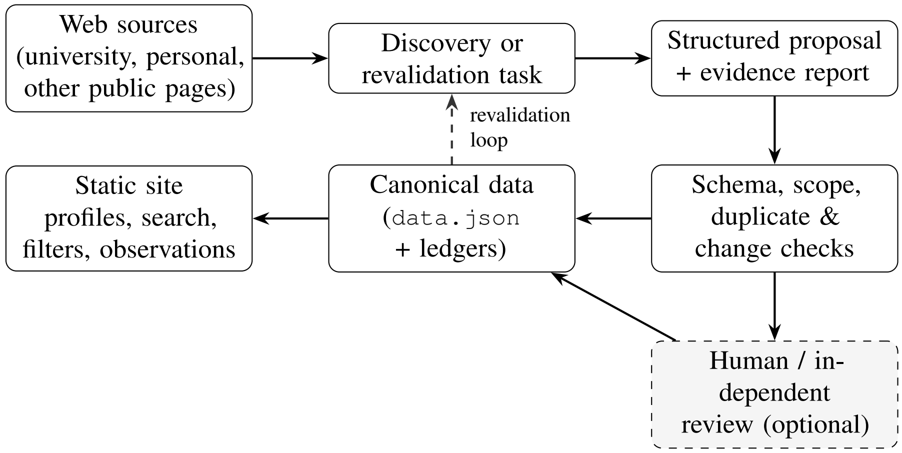

Figure 5: VietProfs architecture. Solid arrows denote implemented data flow; the lower loop represents periodic revalidation.

10 Why Maintaining the Directory Is Hard

The task has five coupled sources of uncertainty. First, it is open-world: the set of eligible people is not enumerated anywhere, so absence from a search result is weak negative evidence. Second, official sources are heterogeneous. A department directory, a university news article, a CV, a personal homepage, and a conference biography have different roles and freshness. Third, academic titles are not globally interchangeable; a research assistant professor can be a career research rank at one institution and a temporary postdoctoral-like role at another. Fourth, identity resolution is hard when names are common, transliterated, reordered, accented, abbreviated, or shared by multiple people. Finally, web pages are mutable: measurements of web decay find that a substantial fraction of pages disappear or change within a few years [3], so a 404 may indicate a moved page or site redesign rather than a departure.

These conditions make a row better understood as a claim such as "person p currently holds appointment a at institution u, supported by source s," with a future revalidation obligation. The database is a materialized view of claims; the sources and review history are what make the view maintainable.

11 System Architecture

Figure 5 summarizes the implemented flow. Public web sources feed either bounded discovery or a targeted revalidation task. An agent returns a structured proposal and report. The controller applies deterministic schema and scope checks before changing the canonical roster. The static site is built from the resulting snapshot, including one ID-addressed profile page per active record; the same snapshot is later selected for revalidation using the ledger. Table 7 maps this flow onto the concrete files that implement each stage.

11.1 Controller and proposal boundary

`scripts/maintain-roster.ts` is the unattended maintenance controller. It normally selects entries missing from the ledger or whose last full review is at least 365 days old (`--stale-days`, default 365), prioritizing older and incomplete records, and processes at most 40 per new run (`--limit`, default 40). Table 8 lists its command-line surface. It supports dry runs, full or capped sweeps (`--all`, `--total`), name- or field-targeting (`--name`, which the agent itself resolves to a canonical roster member or, for a field-like query, to every member of that field), resumable checkpoints, safe stop/status, automatic retry with increasing waits on rate limits, and a choice of research agent (`--agent`), with Claude as the default and Codex selectable when Claude is unavailable or rate-limited.

Table 7: Pipeline inventory: subsystem, implementation, and its automation boundary.

Subsystem	Files	Automation boundary
Discovery queries	`scripts/faculty-discovery-queries.ts`	Deterministic helper; research remains agent/human.
Agent maintenance	`scripts/maintain-roster.ts`	Claude by default; Codex selectable, or usable as an independent review pass.
Proposal validation	Same controller; `test/maintenance-controller.test.ts`	Deterministic schema/scope checks; no agent output is trusted unchecked.
Canonical data	`public/data.json`	Committed JSON, changed only by an approved patch.
Review ledger	`maintenance/verification.json`	Deterministic prioritization of due entries.
Profile identity/pages	`scripts/assign-profile-ids.ts`; `scripts/generate-profile-pages.ts`	Deterministic IDs, static profiles, and sitemap entries.
Classification/search	`src/data.ts`, `src/main.ts`	Runtime deterministic logic in the browser.
Derived view	`buildFunFacts`, `build*Observations`	Deterministic, recomputed, never persisted.
Human submissions	`submit.html`, `src/submit.ts`	Routed to a maintainer; not applied automatically.
Data validation	`scripts/validate-data.ts`	Deterministic; run in `npm test` and CI.
Deployment	`.github/workflows/deploy.yml`	GitHub Actions: test, build, deploy on push to `main`.

Agents cannot edit repository files directly; they can only read, search, and fetch (their tool access is restricted to `Read`, `Glob`, `Grep`, and, for the research stage, `WebSearch`/`WebFetch`), and they return JSON conforming to controller-generated schemas for target matching, research, and optional independent review. The research schema requires `status` (`complete`/`incomplete`), `action` (`keep`/`update`/`remove`), a `proposedEntryJson` string, a free-text `report`, and an array of `sources`; the optional review schema requires a `verdict` (`approve`/`reject`/`uncertain`), a `summary`, `reasons`, and `verifiedSources`. The controller, not the model, is the only component permitted to touch `public/data.json`, and it does so only after `proposalValidationError` returns no error. It preserves a target's immutable ID during an update and deletes an approved removal from the roster. That function—mirrored in `scripts/validate-data.ts` so the same invariants hold whether data changed through the controller or by hand—rejects a proposal that is missing a required string field; whose `profileUrl` is not HTTP(S); whose `websiteUrl` equals `profileUrl` or is not HTTP(S); whose `scholarUrl` is not HTTPS; whose `track` is outside the five accepted values; whose portrait path does not match `^portraits/[a-z0-9][a-z0-9.-]*\.webp$` or whose `portrait`/`portraitSource` are not both present or both absent; whose honors are missing a name/organization/HTTPS source, use an unsupported category (Table 5), have an out-of-range year, or duplicate an existing honor by name/year/organization; or whose degree years fall outside 1900–present. `analyzeRosterProposal` separately diffs the before/after roster and refuses a proposal that reorders the file, inserts an unrelated record, or silently edits a person other than the one under review—an agent that "fixes" something outside its assigned target is treated the same as a malformed proposal. A changed public record receives a new `lastUpdatedAt`; a complete review that finds no substantive change advances only the ledger timestamp, which is how the controller distinguishes "reviewed and unchanged" from "never reviewed" when selecting the next batch. [CODE: `scripts/maintain-roster.ts`; `test/maintenance-controller.test.ts`]

Table 8: Controller CLI flags (`scripts/maintain-roster.ts`).

Flag	Effect
`--limit N`	Entries processed in one new run (default 40).
`--total N`	Process up to N entries across repeated `--limit` batches, committing and pushing after each.
`--all`	Process the entire roster in repeated batches, oldest ledger entries first.
`--stale-days N`	Minimum age, in days, of the last full verification to be selected (default 365).
`--name QUERY`	Ask the agent to resolve a person or field query and queue exactly that target, ignoring normal age/limit selection.
`--dry-run`	Show the selection with no agent call, Git write, commit, or push.
`--codex-review`	Require an independent Codex verdict (approve/reject/uncertain) before applying a proposal.
`--agent A`	Research/propose with `claude` (default) or `codex`.

11.2 Resumable state and unattended operation

The controller keeps its working state—which person and stage are in progress, per-agent logs, and retry counters—outside the repository, by default under `~/.local/state/vietprofs-maintenance/` and overridable with `VIETPROFS_MAINTENANCE_STATE_DIR`. Running `./scripts/maintain-roster.ts run` again, even after the process was killed or the terminal closed, detects the saved checkpoint and resumes the interrupted person and stage rather than restarting the batch; `status` reports progress and `stop` pauses safely. A background invocation (`nohup ./scripts/maintain-roster.ts run &`) can therefore run for the hours a full sweep of the roster takes, including waiting out account rate limits, without an attended terminal. The controller commits and pushes once per completed batch, not once per person, so a batch is the unit of both reviewability and atomicity in the Git history; the first full sweep of the roster was split into 20 batches of 35–40 people to keep individual commits reviewable.

11.3 Independent review and bounded revision

When `--codex-review` is enabled, a second, independently-invoked agent receives the same evidence and the first agent's proposal and returns a verdict rather than a rewrite; the controller applies the proposal only on `approve`. On `reject`, the controller gives the original research agent up to two revision attempts (`MAX_PROPOSAL_REVISIONS = 2`), each carrying the reviewer's concrete reasons, followed by a fresh independent review of the revision; it never applies only the convenient subset of a rejected proposal. A proposal still rejected after its revisions, or a review that returns `uncertain`, is logged, keeps its old verification timestamp, and is deferred for 30 days so a single hard case does not block the rest of a batch. This is the concrete mechanism behind the paper's operational slogan: even with two independent models in the loop, the controller's deterministic checks and the revision/defer policy—not either model's confidence—decide what reaches the public roster.

11.4 Deployment

The deployment workflow (`.github/workflows/deploy.yml`) runs `npm test` and a production build on every push to `main`, then deploys the static artifact to GitHub Pages. The build assigns missing IDs, generates the ID-addressed profile pages, and extends the sitemap with active profile URLs. The maintenance controller can commit and push directly after its own checks pass, so a substantive change reaches the live site without a pull request or manual approval step. This is a strong degree of operational automation, but it is not a formal correctness guarantee and does not eliminate human judgment for ambiguous evidence; Section 15 describes what the test suite actually checks before that push is allowed to happen.

12 Discovery as Partitioned Open-World Search

An instruction such as "find all Vietnamese professors at universities worldwide" has unbounded scope and unmeasurable recall. VietProfs instead generates repeatable queries for a bounded university, field, and domain. `scripts/faculty-discovery-queries.ts` takes `--university`, `--field`, and an optional `--domain` and expands them into a fixed template: site-restricted faculty/professor/directory/news/research-center searches, the same queries repeated for each of a list of common Vietnamese surnames both site-restricted and open, a Google Sites-restricted query for personal/lab pages, and a query aimed at people who "moved from" the institution, to catch departures as well as arrivals. For example, `--university "New Jersey Institute of Technology" --field "Data Science" --domain njit.edu` expands to more than 25 concrete queries such as `site:njit.edu "New Jersey Institute of Technology" "Data Science" faculty` and `site:njit.edu "New Jersey Institute of Technology" "Nguyen" "Data Science"`. The agent is expected to audit existing entries for that slice first, gather evidence per candidate, and treat coauthors, student pages, grants, and research mentions as leads rather than proof.

The surname list itself is not fixed forever: a plain token-frequency scan of already-published records in `public/data.json` surfaced seven common Vietnamese surnames (`Huynh`, `Duong`, `Truong`, `Dang`, `Ngo`, `Mai`, `Dao`) missing from the original ten, simply because they occurred often enough among people already found by other means to be worth adding to the query template. A separate, opt-in mode substitutes common given/middle-name tokens (`Thanh`, `Quang`, `Minh`, and similar) for surnames, restricted to the same institution and paired with explicit exclusion terms (`-Vietnam -student -postdoctoral`) to offset the higher false-positive rate a given-name token carries on its own, since it matches anyone with that token anywhere in their name rather than only as a family name.

Two informal trials of these variants produced different outcomes worth reporting honestly rather than averaging together. The first, run across a handful of otherwise-unremarkable universities in the United States, Japan, and Australia, found six new, independently verified faculty records from a modest number of queries—a hit rate comparable to the original surname-only template—split evenly between a surname missing from the original list and a given-name token; a control run repeating the same technique against a country slice already exhausted by prior search found nothing, weak evidence the six hits reflect the technique rather than an unsearched region. The second trial, testing several further given-name tokens without an institution restriction, found zero new people: three tokens returned only generic academic-rank definitions with no usable lead, and the two tokens that did return a plausible faculty match both turned out to already be in the roster under a different profile URL—a net gain for data completeness (each contributed a previously undocumented fact to its existing record) but not a new addition, and a reminder that an unrestricted given-name search should be checked against the existing roster before anything else, since the token alone narrows down little about which specific person it names. One candidate surname was excluded from the default list on inspection: it is also a common Chinese-Indonesian family name, and its one hit in an unrestricted search was a plausible-looking but non-Vietnamese person. This remains an anecdotal, two-session comparison rather than a controlled evaluation of query-template variants. [CODE: `scripts/faculty-discovery-queries.ts`; DOC: `ROSTER_MAINTENANCE.md`]

This decomposition is a practical form of bounded open-world search. It improves auditability: a research pass can say which university/field/country slice it attempted, what was added, and in some cases that no eligible addition was found—for example, one country-sweep commit for Taiwan records a thorough negative check with no additions, an auditable non-result rather than silence. It does not solve recall. Search-engine ranking, English-language visibility, uneven university websites, and differing effort across slices remain sampling mechanisms. The repository history shows a switch to country-by-country searches, batches for international universities, and field audits. These are operational strategies, not evidence of exhaustive coverage.

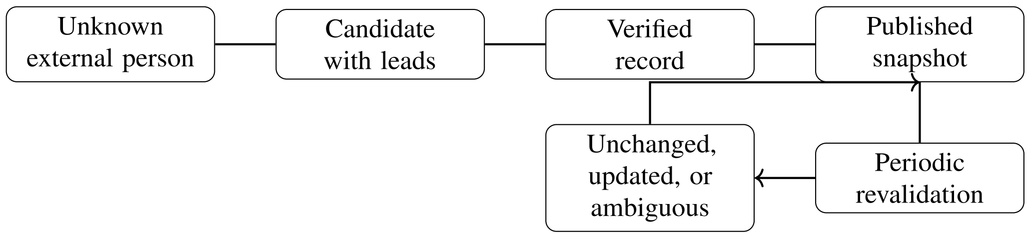

Figure 6: Record lifecycle. A published record is not terminal; it returns to the revalidation loop.

13 Evidence and Verification

Discovery is not verification. A candidate enters the workflow because a query or lead suggests a person; publication requires evidence that the person is current, at the claimed institution and department, in an eligible track, with a working official profile. Official faculty pages are the preferred source. Personal pages and CVs can supply details, but the current appointment is still confirmed on an official source where possible. Search snippets, conference biographies, and coauthorship are leads or corroboration rather than default authority.

The implementation therefore follows the rule "AI proposes; evidence decides," a specific instance of the broader interactive-machine-learning principle that a model's role is to generate candidates and reduce a human reviewer's effort, not to make the final call unsupervised [2]. The agent can read public pages, compare an old record with current evidence, and produce structured data, but the controller does not treat free-form model text as canonical state (Section 11 describes the exact schema and validation gates this passes through). Rejected proposals receive bounded revisions in the configured workflow; incomplete or uncertain work is deferred rather than marked successfully verified. A person's original submission through the public form (Section 7.3) enters this same pipeline as a lead rather than being written directly, so a visitor-supplied link or name is treated with the same research-workflow rigor as an agent-discovered candidate: the maintenance guide directs a reviewer to resolve identity, search the canonical roster for the person first, and independently verify the full inclusion standard before adding or correcting anything, rather than trusting facts merely because they appear at a supplied URL.

The evidence model is practical rather than a formal confidence score. URLs are retained for the profile, honors, portrait, and optional personal/Scholar pages, and Git preserves the resulting patch. There is no per-field provenance graph or source snapshot store. This is an important limitation: a URL is evidence access, not a cryptographic proof that the page had a particular content at review time.

14 Continuous Revalidation and Self-Maintenance

VietProfs separates two loops, shown in Figure 6. Expansion starts from an unsearched slice and tries to add a new verified record. Revalidation starts from an existing record and checks whether its claim still matches the web. The latter checks profile liveness and role, URL-role correctness, Scholar identity, current rank/track, education, honors, and possible moves; it also looks for new candidates incidentally. The ledger allows the controller to choose the least recently reviewed records rather than repeatedly touching only newly added people.

The periodic full-roster refresh workflow specifies six ordered checks per person: (1) confirm `profileUrl` is not dead, a generic not-found page, or a page that no longer identifies that specific person, and if it is, search for a current official profile and either update the record's university/field/rank/track or remove the person with a commit-message reason if they no longer qualify; (2) independently determine which of the stored `profileUrl` and `websiteUrl` is actually the official university page and which is the personal/lab page, and correct the two even when both currently resolve; (3) find and verify a Google Scholar profile using affiliation, research area, and publication corroboration rather than name matching alone; (4) read the official and personal/lab pages fully and update rank/track, education, and honors under the eligibility rules of Section 6.1; (5) treat any Vietnamese-diaspora name surfaced incidentally—a coauthor, a former lab member who became faculty—as a lead for later, not a reason to stall the current batch; and (6) record a successful verification in the ledger only once every applicable check above is complete, even when nothing else changed, so that future batches can reliably select the oldest entries.

A complementary, cheaper revalidation mode targets internal consistency in `public/data.json` rather than external liveness. Two structural gaps are easy to detect without any new search: a degree year stored without its paired institution (for example `undergradYear` present but `undergradInstitution` absent), and a record with no education field populated at all despite an already-vetted `profileUrl`. Both are found by a plain scan of the roster file, and both are resolved, when they can be, by re-fetching only that person's existing official page rather than launching a discovery query. Because the source is already known and was already accepted once, this check skips the liveness, URL-role, and Scholar-identity steps above and asks only whether the previously reviewed page in fact states the missing fact. The two gap types differ sharply in yield: a year-without-institution mismatch is close to always resolvable, since the page that supplied the year almost always names the institution nearby; a record with no education fields at all resolves at a much lower rate, since directory-style listings for adjunct, lecturer, and teaching-track staff frequently omit degree history entirely, while structured CV pages at academic medical centers (which routinely publish a dedicated education section) are disproportionately productive. A sweep of the latter kind should report the specific records left unresolved rather than treat a silent page as evidence that no degree exists.

Negative evidence requires caution. A disappeared faculty page can mean redesign, a moved URL, temporary failure, a university move, retirement, or departure. The maintenance policy explicitly avoids treating `404` as automatic deletion and calls for corroboration. Removal is also a stronger action than addition because it destroys a public claim and can erase a legitimate person from the directory. The system supports removal proposals, but the guide's corroboration rule is an operational policy rather than a formally verified deletion protocol. A confirmed removal deletes the record and its generated profile page.

The Git history makes this lifecycle visible. A 20-batch full-roster refresh appears on 26 August 2026. Subsequent history records automated maintenance batches, country sweeps for Australia, Canada, France, Singapore, Taiwan, Japan, Germany, and Hong Kong among others, a Teaching-to-Clinical correction, a duplicate merge, a university-name canonicalization, a corrected affiliation, and a policy-based removal. These examples show that maintenance is not a periodic cosmetic edit; it changes eligibility, identity, affiliation, and derived data assumptions.

15 Testing and Validation

Because the controller is authorized to push directly to the branch that GitHub Actions deploys, the test suite that gates it is load-bearing rather than advisory. `npm test` runs `scripts/validate-data.ts` followed by four Node test files. `validate-data.ts` independently re-implements the roster invariants also enforced at proposal time—required-field presence, unique immutable IDs, canonical UTC `lastUpdatedAt`/`lastVerifiedAt` timestamps that are not in the future, HTTP(S)/HTTPS URL roles, portrait path shape and on-disk existence, honor category and duplicate checks, degree-year ranges, accepted tracks, non-empty `researchAreas`—and cross-checks `public/data.json` against `maintenance/verification.json`: every canonical name must have exactly one ledger entry, and the ledger must not contain an entry for a name no longer on the roster. `test/data.test.ts` covers the field-mapping taxonomy and its overrides, the diacritic-insensitive search index, and the derived-facts builders in Section 8, including their behavior on small or empty filtered rosters. `test/maintenance-controller.test.ts` exercises `proposalValidationError` and `analyzeRosterProposal` directly against constructed before/after rosters, including the scope-violation cases (reordering, inserting an unrelated person, editing outside the target) that Section 11 describes. `test/profile-pages.test.ts` verifies unique IDs and the generated profile route contract, including a name correction preserving its ID-addressed URL. `test/ui.test.ts` checks rendering logic such as card content and leaderboard construction. A separate `npm run test:e2e` runs `test/browser-smoke.test.ts` with Playwright against a built site, loading the real `index.html` and `submit.html` in a headless browser to catch integration failures—a broken search box, a filter

Table 9: Representative public maintenance events in Git history.

Case	Repository evidence	Result
Duplicate	`2aa15c0`; University of Dayton	Duplicate entries for Tam V. Nguyen merged.
Policy removal	`9a4a171`; inclusion-standard decision	Nhung Nguyen (UCSF) removed.
Track correction	`ba71beb`; BU Dental	Teaching changed to Clinical.
Affiliation correction	`c1f4380`	Nguyen-Truc-Dao Nguyen affiliation corrected.
Name/identity cleanup	`c0c0603`, `4212955`	True duplicate, suffix, and university-name canonicalization fixed.
Negative discovery	`9b31b95`; Taiwan country sweep	Thorough search, no eligible addition found.

that does not update the URL, a submission form that cannot find an existing record—that unit tests over the data layer cannot see. `npm run typecheck` runs the TypeScript compiler with no emit as a fourth, independent check. The maintenance guide additionally requires `npm test`, `npm run build`, and `git diff --check` before any roster commit, whether made by a human or the controller.

16 Operational Experience and Evaluation

This work is an operational characterization, not a controlled experiment. The repository provides strong evidence about implementation and history but not precision, recall, cost, or detection latency. The current snapshot has 774 Tenure-line, 96 Clinical, 68 Teaching, 37 Emeritus, and 18 Research records. It contains 127 personal/lab URLs, 219 Scholar URLs, and 182 records with at least one honor. Of the 751 records with an explicit PhD institution, PhD years span 1962–2026 across 549 records with an explicit year, a wide enough range that any framing of the roster as capturing only recent arrivals in the diaspora would be unsupported. These figures describe fields present in the snapshot, not their correctness or completeness.

The event history nevertheless supports meaningful engineering observations. Batch commits make large work reviewable; checkpoints make long agent sessions resumable; tests encode invariants such as unique names, unique profiles, valid tracks, and targeted proposal scope; and the ledger separates "reviewed" from "changed." The absence of candidate-level telemetry is equally important. There is currently no denominator for candidates considered, no rejection taxonomy, no measured human review rate, and no time-to-detection metric; Section 17 and the manuscript notes in the repository (`PAPER_NOTES.md`) enumerate what a controlled evaluation would need to add: per-run candidate outcomes, rejection reasons, elapsed time and token/API cost per person, and a measured interval between an external page changing and the directory detecting it.

17 What Does Not Work

Unbounded LLM enumeration is a poor source of a canonical roster. It omits people, hallucinates affiliations—a documented failure mode of language-model generation in general, not one specific to this task [7]—and cannot provide a defensible stopping condition. Surname inference is unsafe and ethically inadequate [10, 1]. Search snippets are stale and incomplete. Direct model writes would make it difficult to constrain scope, distinguish a proposed change from an approved one, or recover from a mistaken bulk edit. VietProfs instead requires structured proposals and targeted changes.

Other failures are subtler. A broad title such as "Research Assistant Professor" cannot establish eligibility by itself. A page disappearance is not a departure. A personal homepage is not the same source role as an official university profile, and the two are easy for either a human or a model to swap even when both currently resolve, which is why the periodic-refresh workflow checks both roles independently rather than trusting whichever URL happens to already be stored in each field. Department keywords misclassify ambiguous institutional contexts unless overrides are maintained. Finally, a current count cannot be described as growth or a migration path without versioned historical data and an explicit comparison. These lessons are implemented in policy, tests, and recent corrective commits, but they remain areas for ongoing measurement.

18 General Lessons

The case study suggests the following design principles for agent-maintained public datasets:

1. Use language models primarily for candidate generation, comparison, and research assistance; do not make them the source of truth.
2. Separate discovery, evidence collection, verification, proposal, and canonical application.
3. Attach source URLs and review timestamps to claims, while recognizing that URLs alone are not complete provenance.
4. Partition open-world search into bounded, repeatable tasks.
5. Treat removal and negative evidence as higher-risk than addition.
6. Keep candidate state separate from canonical state and constrain proposed edits to a target.
7. Reserve human attention for ambiguity, identity resolution, policy exceptions, and disputes.
8. Design telemetry before deployment: outcomes, reasons, cost, latency, and reviewer actions.
9. Recompute or invalidate derived knowledge when primary claims change.
10. Route human-submitted leads through the same evidence pipeline as agent-discovered ones, rather than trusting a visitor-supplied fact merely because a person asked for it.

Together these principles implement the operational slogan "AI proposes; evidence decides" without pretending that the web or the model is fully reliable.

19 Related Work

VietProfs is related to entity resolution, web extraction, scholarly knowledge graphs, and provenance-aware maintenance, but it is not a replacement for a general scholarly index. Surveys of entity resolution emphasize end-to-end blocking, matching, and data-quality pipelines over heterogeneous descriptions [5, 6]. Recent work studies LLMs for entity matching, including prompt sensitivity, structured explanations, and robustness to unseen entities [15]; another line explicitly treats model queries as a costed uncertainty-reduction resource [9]. VietProfs uses the practical implication—models can help compare and organize evidence—but does not claim their reported benchmark performance for this domain.

Knowledge-graph maintenance work formalizes incremental updates and provenance for asserted and derived facts [4]. VietProfs is simpler: JSON records, URLs, Git history, and runtime aggregation rather than an RDF graph or why-provenance engine. The conceptual connection is nevertheless direct: when a source claim changes, dependent answers may change.

OpenAlex provides a useful contrast as a large, open scholarly knowledge graph of works, authors, venues, institutions, and concepts [16]. VietProfs has a narrower population, more detailed appointment eligibility, and a public-directory interaction model. DBLP and ORCID are also important reference ecosystems for scholarly identity, but VietProfs's engineering problem is the current faculty appointment and its evidence, not publication identity alone. The classification literature underlying Section 4—the NCES Classification of Instructional Programs and the NSF Survey of Earned Doctorates taxonomy [12, 13], and the NIH's Institute/Center structure [14]—is federal statistical and administrative infrastructure rather than research literature, but it is what lets VietProfs's field taxonomy be read against national context instead of standing alone. The contribution here is therefore twofold: a public, purpose-built directory for a specific and previously unserved community, and an experience report about the operational boundary between agent assistance and canonical data in a specialized, continuously revalidated directory.

20 Conclusion

A directory like VietProfs is not a database to be finished; it is a relationship to be kept. The contribution of this paper is not that we solved data-quality engineering in general. It is that a small, name-independent inclusion rule, paired with an evidence-gated agent pipeline, let a resource this community never had exist and stay trustworthy, without demanding a permanent human maintenance staff. That is worth having on its own terms. A graduate student can now find an advisor who shares her background in one search instead of dozens. A professor moving countries can see who is already there. A community that spans continents and six decades—from Xuong Nguyen-Huu's 1962 PhD to next year's newest hire—can, for the first time, see itself in one place.

We do not think this is the end state, and a few directions we are actively considering would turn VietProfs from a static listing into something closer to a living network. One is opt-in academic genealogy: letting advisors and their students link their entries, so a visitor could trace not just who is where but who trained whom, the way scholars already trace mathematical and scientific lineages, revealing multi-generation chains of Vietnamese-diaspora mentorship that no single record shows on its own. Another is proximity and mentorship matching: surfacing "N Vietnamese-diaspora faculty within 50 miles of you," or letting a prospective student opt in to informal mentorship with a faculty member outside their own institution, turning a search result into an introduction. A third, closer to the scientific-diaspora literature that motivated this project (Section 1.1), is a deliberate bridge back to Vietnam: letting universities and researchers inside the country discover and reach out to diaspora faculty for guest lectures, joint appointments, and collaboration, so the directory serves circulation between two academic communities rather than only enumeration of one. A fourth is self-verification: letting the people listed claim, correct, and extend their own entries directly, turning today's one-directional maintenance pipeline into a two-way one where the subjects of the directory become its co-maintainers.

None of this requires abandoning the boundary this paper argues for. A model can propose a genealogy edge, a mentorship match, or a self-submitted correction, but a deterministic gate—and, where needed, a human—should still decide what becomes a public claim. What changes is the ambition. The engineering pattern here, agents propose and evidence decides while a small team keeps a few hundred records honest against a changing web, is only useful in service of a larger one: giving a diaspora that never had a single place to find itself the beginning of one, and leaving room for that place to grow into something people actually use to find each other, not merely to be found.

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